

Jonathan Nicholas, Ph.D.

✉ jdmnichol@gmail.com 🐦 @jonathannicholas.bsky.social
🌐 <https://jonathanicholas.github.io/>

Employment History

- 2023 – now **Postdoctoral Researcher**, New York University
Advisor: Marcelo Mattar
- 2015 – 2017 **Research Software Developer**, Stanford Cognitive and Systems Neuroscience Lab

Education

- 2017 – 2023 **Ph.D., Columbia University**, Psychology (Cognitive Neuroscience)
Advisor: Daphna Shohamy
- 2011 – 2015 **B.Sc. with Honors, Brown University**, Cognitive Neuroscience

Grants and Fellowships

- 2025-2027 NSF SBE Postdoctoral Research Fellowship, \$160,000
"Exploring the role of episodic memory in adaptive decision making"
- 2021-2022 Leo Rubinstein Endowed Fellowship
- 2017-2020 NSF Graduate Research Fellowship

Honors and Awards

- 2024 Best Talk Award, Society for Neuroeconomics
Postdoctoral Travel Award, New York University
- 2022 Edward E. Smith Memorial Award in Cognitive Neuroscience, Columbia University
- 2015 Kling Premium in Psychology, Brown University
- 2014 Karen T. Romer Undergraduate Teaching and Research Award, Brown University
1st Prize, Brown Institute for Brain Sciences Neural Decoding Competition

Research

Journal Articles

- 1 **Nicholas, J.** and Mattar, M.G., "Episodic memory facilitates flexible decision making via access to detailed events," *Nature Human Behaviour*, 2026. 🌐 DOI: 10.1101/2025.03.13.643066.
- 2 Lichtman, D., Bergmann, E., **Nicholas, J.**, Gerraty, R.T., and Kahn, I., "The periaqueductal gray selectively supports reversal learning during a flexible discrimination task in mice," *eLife*, 2026. 🌐 DOI: 10.64898/2026.01.19.700312.
- 3 **Nicholas, J.**, Daw, N.D., and Shohamy, D., "Proactive and reactive construction of memory-based preferences," *Nature Communications*, 2025. 🌐 DOI: 10.1038/s41467-025-56183-4.
- 4 Montaser-Kouhsari, L.*, **Nicholas, J.***, Gerraty, R.T., and Shohamy, D., "Differentiating reinforcement learning and episodic memory in value-based decisions in parkinson's disease," *Journal of Neuroscience*, 2025, ***Denotes co-first author.** 🌐 DOI: 10.1523/JNEUROSCI.0911-24.2025.

- 5 **Nicholas, J.**, Amlang, C.J., Lin, C.Y., Desai, N., Montaser-Kouhsari, L., Kuo, S.H., and Shohamy, D., "The role of the cerebellum in learning to predict reward: Evidence from cerebellar ataxia," *The Cerebellum*, 2024. [DOI: 10.1007/s12311-023-01633-2](#).
- 6 Luo, X., Rechard, A., Sun, G., N. Yanez, F., Yilmaz, B., Lee, K., Cohen, A.O, Borghesani, V., Pashkov, A., Marinazzo, D., **Nicholas, J.**, ..., and Love, B.C., "Large language models surpass human experts in predicting neuroscience results," *Nature Human Behaviour*, 2024. [DOI: 10.1038/s41562-024-02046-9](#).
- 7 **Nicholas, J.**, Daw, N.D., and Shohamy, D., "Uncertainty alters the balance between incremental learning and episodic memory," *eLife*, 2022. [DOI: 10.7554/eLife.81679](#).
- 8 Grossman, I., Rotella, A., Hutcherson, C.A., ..., **Nicholas, J.**, ..., and Wilkening, T., "Insights into accuracy of social scientists' forecasts of societal change," *Nature Human Behaviour*, 2022. [DOI: 10.1038/s41562-022-01517-1](#).
- 9 Chen, L., Iuculano, T., Mistry, P., **Nicholas, J.**, Zhang, Y., and Menon, V., "Linear and nonlinear profiles of weak behavioral and neural differentiation between numerical operations in children with math learning difficulties," *Neuropsychologia*, 2021. [DOI: 10.1016/j.neuropsychologia.2021.107977](#).
- 10 Iuculano, T., Padmanabhan, A., Chen, L., **Nicholas, J.**, Mitsven, S., de los Angeles, C., and Menon, V., "Neural correlates of cognitive variability in childhood autism and relation to heterogeneity in decision-making dynamics," *Developmental Cognitive Neuroscience*, 2020. [DOI: 10.1016/j.dcn.2020.100754](#).
- 11 Dimsdale-Zucker, H.* and **Nicholas, J.***, "Is spatial context privileged in the neural representation of events?" *Journal of Neuroscience*, 2018, *Denotes co-first author. [DOI: 10.1523/JNEUROSCI.0949-18.2018](#).
- 12 Taghia, J., Cai, W., Ryali, S., Kochalka, J., **Nicholas, J.**, Chen, T., and Menon, V., "Uncovering hidden brain state dynamics that regulate performance and decision-making during cognition," *Nature Communications*, 2018. [DOI: 10.1038/s41467-018-04723-6](#).
- 13 Ryali, S., Supekar, K., Chen, T., Kochalka, J., Cai, W., **Nicholas, J.**, Padmanabhan, A., and Menon, V., "Temporal dynamics and developmental maturation of salience, default and central-executive network interactions revealed by variational bayes hidden markov modeling," *PLOS Computational Biology*, 2016. [DOI: 10.1371/journal.pcbi.1005138](#).

Conference Proceedings

- 1 **Nicholas, J.** and Mattar, M.G., "Humans use episodic memory to access features of past experience for flexible decision making," in *46th Proceedings of the Annual Meeting of the Cognitive Science Society*, Rotterdam, The Netherlands, 2024. [URL: https://escholarship.org/uc/item/9x22d800](https://escholarship.org/uc/item/9x22d800).

Presentations

Invited and Selected Talks

- | | |
|------|--|
| 2025 | The University of Texas at Austin Department of Psychology Special Seminar , Austin, TX
UCL Max Planck Computational Psychiatry Seminar Series , Virtual
Johns Hopkins Ocular Motor & Vestibular Lecture Series , Virtual |
| 2024 | Annual Meeting of the Cognitive Science Society , Rotterdam, Netherlands
Society for Neuroscience , Chicago, IL
Society for Neuroeconomics , Cascais, Portugal |
| 2022 | 5th Multidisciplinary Conference on Reinforcement Learning and Decision Making , Providence, RI
Society for Neuroeconomics , Arlington, VA |

Presentations (continued)

- 2019 **Columbia Interdisciplinary Decision Making Meeting**, New York, NY
 Manhattan Area Memory Meeting, Princeton, NJ

Posters

- 2025 **6th Multidisciplinary Conference on Reinforcement Learning and Decision Making**,
 Dublin, Ireland
- 2024 **Cognitive Computational Neuroscience**, Boston, MA
- 2022 **Society for Neuroscience**, San Diego, CA
 Neurobiology of Reward and Decision Making, Lake Arrowhead, CA
 18th Annual Context and Episodic Memory Symposium, Philadelphia, PA
 International Congress of Parkinson's Disease and Movement Disorders, Madrid, Spain
 International Congress for Ataxia Research, Dallas, TX
- 2019 **Society for Neuroscience**, Chicago, IL
 Cognitive Neuroscience Society, San Francisco, CA
- 2016 **Fourth Annual Flux Congress**, St. Louis, MO
- 2014 **Brown Summer Research Symposium**, Providence, RI

Academic Service

Organizing

- 2025 **Application Support Mentor**, NYU Application Support Group
- 2025-2026 **Organizer**, NYU ConCats Seminar Series
- 2024-2025 **Organizer**, NYU ConCats Seminar Series
- 2022 **Research Mentor**, Columbia Summer Internship Program in Psychological Science
- 2020-2021 **Instructor**, Columbia University Introduction to Programming Bootcamp
- 2020 **Organizer**, Columbia Interdisciplinary Decision Making Meeting
 Scientific Computing Support Staff, Columbia Psychology Department
- 2019 **Organizer**, Columbia University Introduction to Programming Bootcamp
 Organizer, Manhattan Area Memory Meeting

Teaching

- 2022 **Teaching Fellow**, Science of Psychology, Columbia University
 Teaching Fellow, Cognitive Neuroscience, Columbia University
- 2020 **Teaching Fellow**, Statistics for Behavioral Scientists, Columbia University
- 2019 **Teaching Fellow**, Cognitive Neuroscience, Columbia University
- 2018 **Teaching Fellow**, Experimental Methods, Columbia University
- 2017 **Teaching Fellow**, Behavioral Neuroscience, Columbia University
- 2015 **Teaching Assistant**, Computational Cognitive Science, Brown University

Mentoring

Undergraduate Honors Theses

Academic Service (continued)

2019-2020 Nicole van Amerongen

2018-2020 Jessica Hecht

Undergraduate Research Assistants

2025-2026 Dariush Nashat

2023-2025 Yifei Deng

2022 Annie Xu

Sukriti Gupta

2020 Jesse Eiseman

2019 Natasha King

High School Research Assistants

2022 Andy Feng

Hitomi Nakamura

2021-2022 Pradnya Rajalakshmi

2021 Loc Nguyen

Brad Ji

Peer Review

Nature Communications, eLife, PLOS Computational Biology, Journal of Experimental Psychology: General, Cognitive Science, Memory and Cognition, Scientific Reports, Frontiers in Neuroscience, PLOS One

Technical Skills

Research Methods

Task design, Bayesian modeling, Reinforcement learning, Eyetracking, fMRI, MEG

Programming Languages

Proficient: Python, Matlab, Javascript

Competent: Stan, R, Unity/C, Bash, HTML/CSS